

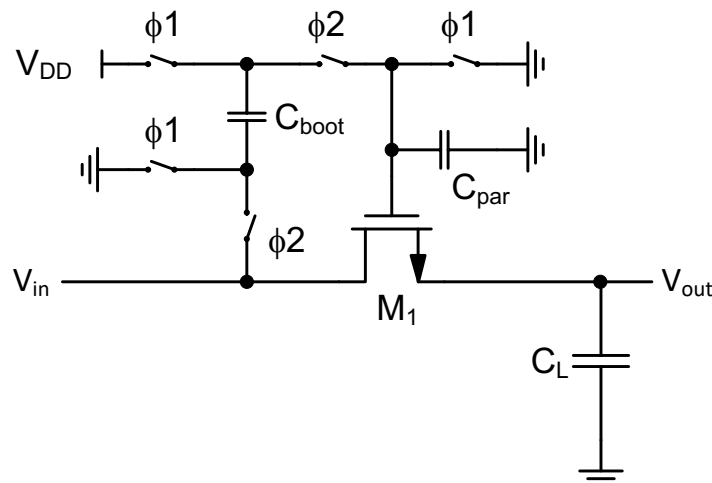
HOMEWORK #3

(Due: Monday, March 13, 2023, 10:30 am via Brightspace)

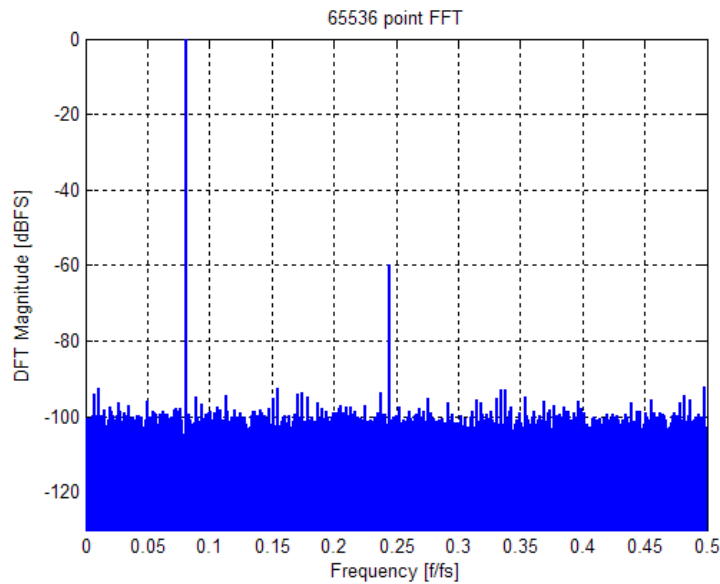
1. **Spectral Performance Metrics.** Download from the website the file `hw3_adc_out.mat`. It contains the output of an ADC with a sampling frequency $f_s = 100 \text{ MS/s}$. The input was a full-scale sinusoidal signal. The ADC has a 10-bit bipolar mid-rise quantizer whose full scale range is $[-1, +1]$.
 - (a) Plot the spectrum from 0 to $f_s/2$. The y axis should be in dBFS while the x axis should be in MHz.
 - (b) What is the frequency of the sinusoidal signal at the input of the ADC, f_{in} ?
 - (c) Compute the following metrics: SNR, SNDR, ENOB, THD, and SFDR.
 - (d) Which non-ideality determines the SFDR in this particular case?
 - (e) Can you tell whether we have other sources of noise in the ADC, in addition to quantization noise? If we do, can you tell how their total power relates to the power of the noise introduced by quantization?
2. **Clock Bootstrapping.** In the sampling circuit below, the input voltage varies between 0.5V and 1V. Ignore all capacitances except the ones explicitly shown. Model the MOSFET using ideal square law expressions and neglect the backgate effect.

Parameters: $C_{boot} = C_L = 500\text{fF}$, $C_{par} = 100\text{fF}$, $V_{DD} = 1.8\text{V}$, $V_t = 0.5\text{V}$, $WLC_{ox} = 50 \text{ fF}$, $\mu C_{ox} W/L = 10\text{mA/V}^2$

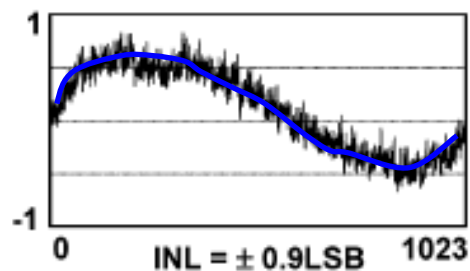
- (a) Calculate the percent increase in M_1 's on-resistance across the input voltage range (relative to its minimum value).
- (b) For the given input voltage range, calculate the peak-to-peak error variation in the sampled output voltage due to charge injection. Neglect clock feedthrough ($C_{ol} = 0$) and assume that the channel charge splits 50/50 (fast gating).



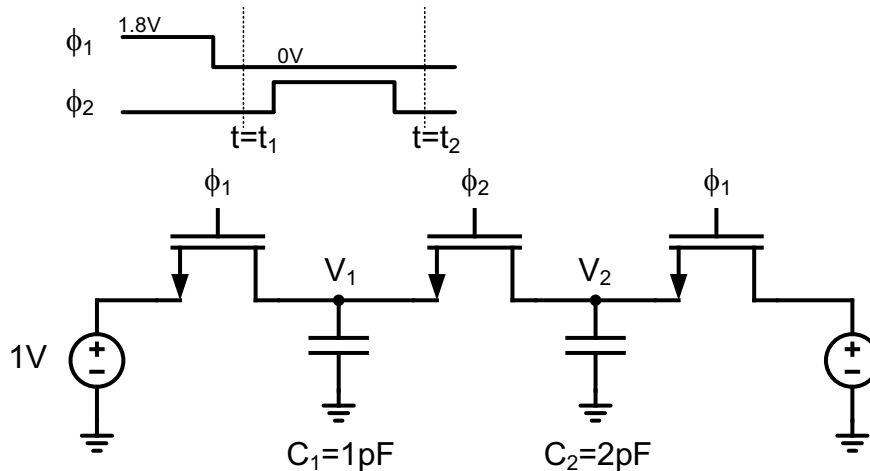
3. **Relationship between INL and SFDR.** Shown below is the output spectrum of a 11-bit A/D converter for a full-scale sinusoidal input.
 - (a) Estimate the maximum INL of the converter in units of LSB. Assume that the INL has a cubic bow.



- (b) As discussed in class, the noise floor of the spectrum is given by the sum of quantization noise and electronic noise. For the above given spectrum and ADC, estimate the standard deviation of the electronic noise component in units of LSB.
- (c) Estimate the ADC's "effective" number of bits.
- (d) Relationship between INL and SFDR in a commercial product. Download the following paper: **Seung-Chul Lee et al. "A 10-bit 205-MS/s 1.0-mm² 90-nm CMOS Pipeline ADC for Flat Panel Display Applications," IEEE J. Solid State Circuits, Vol. 42, No. 12, pp. 2688-2695, Dec. 2007.** The typical INL plot for 1Vp-p single ended input [Fig.14 (b)] shows a dominant cubic nonlinearity. Estimate the converter's SFDR from the INL curve (use the amplitude of the "smooth" average to this curve as suggested by the figure below). Does this number agree with the specified typical SFDR value (use the typical value at 70 MHz input frequency)?



4. **Switch nonidealities.** In the circuit below, all switches are identical NMOS devices with $W=10\mu\text{m}$, $L=0.2\mu\text{m}$, $C_{ox}=10\text{fF}/\mu\text{m}^2$, $V_t=0.4\text{V}$, $C_{ol}=0.1\text{fF}/\mu\text{m}$. Ignore settling dynamics time in this problem (assume that the clock period is much longer than the settling time for each transition) and assume fast gating.



- (e) Calculate the standard deviation of V_1 at times t_1 and t_2 . Express your answer in μV_{rms} .
 - (f) Calculate the value of V_1 at time t_1 , taking only charge injection into account (neglect clock feed-through).
 - (g) Calculate the value of V_1 at time t_1 , taking only clock feed-through into account.
5. **Boxcar Sampling.** Read the first four pages of this paper:
H. Gao, R.M. Walker, P. Nuyujukian, K.A.A. Makinwa, K.V. Shenoy, B. Murmann and T.H. Meng, "HermesE: A 96-Channel Full Data Rate Direct Neural Interface in 0.13 μm CMOS," IEEE J. Solid-State Circuits, pp. 1043-1055, Apr. 2012. <http://ieeexplore.ieee.org/xpl/articleDetails.jsp?arnumber=6158616>
- (h) Derive equation (2).
 - (i) Verify the 3.4x advantage in the effective noise bandwidth, claimed by the authors.